

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276651

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Hsu; Wei K.

VEHICLE SEAT STORAGE SYSTEM

Abstract

A vehicle seat-top storage device has a lower layer formed from a flexible material configured to overlay and conform to portions of a vehicle seat. An upper layer of a flexible sheet material is coupled along side edges to the lower layer to define an interior space between the lower and upper layers. An opening of the device allows access to the interior space through the opening. A closure device is provided for selectively opening and closing the opening.

Inventors: Hsu; Wei K. (Parker, TX)

Applicant: Hsu; Wei K. (Parker, TX)

Family ID: 95157700

Appl. No.: 19/070866

Filed: March 05, 2025

Related U.S. Application Data

parent US continuation 18761470 20240702 parent-grant-document US 12263813 child US 19070866

parent US continuation-in-part 18593313 20240301 parent-grant-document US 12024129 child US 18761470

Publication Classification

Int. Cl.: B60R7/04 (20060101); B60R11/00 (20060101); B60R11/02 (20060101)

U.S. Cl.:

CPC B60R7/043 (20130101); B60R2011/0015 (20130101); B60R2011/0059 (20130101); B60R2011/0075 (20130101); B60R2011/0082 (20130101); B60R11/0241 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/761,470, filed Jul. 2, 2024, which is a continuation-in-part of U.S. patent application Ser. No. 18/593,313, filed Mar. 1, 2024, now U.S. Pat. No. 12,024,129, each of which is incorporated herein by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to storage devices and seat covers for vehicles.

BACKGROUND

[0003] While vehicle seats serve their primary function of providing seating for vehicle occupants, there is a need for an improved storage system that optimizes the utilization of the available space of unoccupied vehicle seats to be used to store and retain items on the seat so that they are readily accessible and that may provide protection to the seat itself.

[0004] Current solutions for in-vehicle storage vary in design and functionality, including traditional glove compartments, center consoles, seatback pockets, etc. These existing storage options may be less convenient or unsuitable for storing certain items. While vehicle occupants may at times place bags or other articles on the seat itself, this can damage or soil the seat. Additionally, abrupt or sudden stops of the vehicle may cause the items to slide off the seat top to a position where they are not easily retrieved or that may result in damage to the item.

[0005] The present disclosure describes ways to overcome these limitations and contributes to the efficient use of space within vehicles by providing a seat-top storage device that is specifically designed for a vehicle seat. The device offers a secure and organized means of storing various items, such as electronic devices, mobile phones, tablets, laptop computers, bags or purses, shopping bags, clothing items, documents, food items and snacks, or other personal items that are easily stored, readily accessible and prevented from sliding off the seat. Additionally, the device is designed for easy positioning and securing to the vehicle seat, as well as easy detachment, collapsing and storing the device out of sight, so that a passenger may still use the seat in a conventional manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For a more complete understanding of the embodiments described herein, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying figures, in which:

[0007] FIG. 1 is an exploded, perspective view of a seat-top storage device constructed in accordance with particular embodiments of the disclosure;

[0008] FIG. 2 is a top plan view of the seat-top storage device of FIG. 1, shown assembled with an upper opening of the device in a closed condition;

[0009] FIG. 3 is a left front perspective view of the seat-top storage device of FIG. 2, shown employed on a vehicle passenger seat with the upper opening in an open condition;

[0010] FIG. 4 is a left-side elevational view of the seat-top storage device and vehicle passenger seat of FIG. 3, with an upper opening in the closed condition;

[0011] FIG. 5 is a rear elevational view of the vehicle passenger seat of FIG. 3, shown with a seat-base securing strap of the seat-top storage device secured to the vehicle seat for securing a lower portion of the device to the vehicle passenger seat;

[0012] FIG. 6 is a left front perspective view of the seat-top storage device of FIG. 3, shown employed on the vehicle passenger seat, with the upper opening in a closed condition

[0013] FIG. 7 is a left front perspective view of the seat-top storage device of FIG. 3, shown employed on the vehicle passenger seat illustrating how items can be stored on top of the device in one mode of operation;

[0014] FIG. 8 is a left front perspective view of the seat-top storage device of FIG. 3, shown employed on the vehicle passenger seat with the upper opening of the device being in an open condition so that an article can be placed in an interior space of the device in another mode of operation;

[0015] FIG. 9 is a left front perspective view of the seat-top storage device and vehicle passenger seat of FIG. 8, shown with the article positioned within the interior space and the upper opening in a closed condition;

[0016] FIG. 10 is an exploded, perspective view of another embodiment of seat-top storage device constructed in accordance with particular embodiments of the disclosure;

[0017] FIG. 11 is a right front perspective view of the seat-top storage device of FIG. 10, shown employed on a vehicle passenger seat with the side opening in an open condition;

[0018] FIG. 12 is a right front perspective view of the seat-top storage device and vehicle passenger seat of FIG. 11, with an upper opening in the open condition and the side opening in a closed condition;

[0019] FIG. 13 is a right front perspective view of the seat-top storage device and vehicle passenger seat of FIG. 11, with an upper opening and side opening in closed conditions with items stored within an interior of the storage device;

[0020] FIG. 14 is a right side elevational view of the seat-top storage device of FIG. 11, shown employed on the vehicle passenger seat, with a pet secured in the storage device; and

[0021] FIG. 15 is a right front perspective view of the seat-top storage device of FIG. 11, shown in a collapsed configuration for storage under a vehicle seat;

[0022] FIG. 16 is a right front perspective view of the seat-top storage device of FIG. 11, shown in a collapsed configuration with ends of the collapsed storage device folded inward for storage under a vehicle seat;

[0023] FIG. 17 is a right front perspective view of the seat-top storage device of FIG. 11, shown in a collapsed configuration with ends of the collapsed storage device folded inward for storage in a storage compartment;

[0024] FIG. 18 is a right front perspective view of another embodiment of a seat-top storage device constructed in accordance with particular embodiments of the disclosure that employs a zipper assembly; and

[0025] FIG. 19 is an enlarged view of the zipper assembly of the seat-top storage device of FIG. 18.

DETAILED DESCRIPTION

[0026] Referring to FIG. 1, a vehicle seat-top storage device 10 is shown. The device 10 is configured as a flexible apron or cover for a vehicle seat and comprises a lower layer 12 of a flexible sheet material that is configured to overlay and conform to portions of the vehicle seat. As shown, the lower layer 12 is defined by left and right side edges 14, 16 that are joined by top and bottom edges, 18, 20. The lower layer 12 includes a base portion 22, configured to overlay a seat base of a vehicle seat, and a backrest portion 24, configured to overlay a backrest of a vehicle seat. In certain embodiments, the lower layer 12 may taper in width along side edges 14, 16 from an area of the base portion 22 to a narrower top end 26 of the backrest portion 24 that terminates at top edge 20.

[0027] All or portions of the lower layer 12 may be a flexible fabric material or non-fabric material or a combination of such materials. The layer 12 may be formed as a single continuous piece of such material or may be formed from several pieces of such materials that are joined together to form a continuous piece. The fabric material may be a woven or non-woven fabric material. Non-limiting examples of fabric materials suitable for the layer 12 include manmade or polymeric

fabrics, such as polyester, polyamide, nylon, aramid, polypropylene, polyethylene, high-density polyethylene (HDPE), low-density polyethylene (LDPE), polyurethane, acrylic, vinyl, PVC, rayon, spandex, fiberglass, carbon fiber, etc. Natural fabric materials may also be used, such as cotton, wool, silk, linen, hemp, bamboo, etc. Non-limiting examples of non-fabric materials for the layer **12** may include flexible polymeric sheet materials such as polypropylene, polyethylene, HDPE, LDPE, vinyl, PVC, silicone, neoprene, foam, etc. A combination of different fabrics and/or non-fabric materials may also be used for all or portions of the lower layer **12**.

[0028] All or portions of the lower layer **12** may be non-elastic or non-stretchable. In other embodiments, however, all or portions of layer **12** may be elastic or stretchable. Additionally, all or portions of the lower layer **12** may be liquid impervious or liquid resistant so that liquids cannot pass or readily seep through the material. This may include liquid-impermeable or liquid-resistant fabrics materials or liquid-permeable fabric materials that have been treated or coated with a liquid impervious or liquid resistant material so that the fabric material is liquid impervious or liquid resistant. Non-limiting examples of suitable liquid-impervious or liquid-resistant fabric materials, coatings and treatment materials for the flexible lower layer **12** may include nylon, synthetic or natural rubber, neoprene, Gore-Tex® or expanded polytetrafluoroethylene (ePTFE) fabric, vinyl, PVC, silicone, wax, acrylic, fluoropolymer, oilcloth, polyurethane, etc. In other embodiments, all or portions of the lower layer **12** may be liquid permeable. All or portions of the upper surface of the lower layer **12**, particularly the base portion **22**, may be texturized, coated or be otherwise provided with a non-slip surface or surface areas. This non-slip surface may be formed from the material of the lower layer **12** itself or as a non-slip layer or coating, such as a rubber, silicone, polyurethane, vinyl, etc., which may have non-slip properties themselves or that are texturized or embossed to have such non-slip properties. Non-slip tape, strips, or layers may also be applied to the surfaces of the lower layer **12** to provide the non-slip properties.

[0029] The device **10** further comprises an upper layer **28** of a flexible material that is defined by left and right side edges **30**, **32** that are joined by top and bottom edges **34**, **36**. The upper layer **28** is coupled along its side edges **30**, **32**, **34**, **36** to the lower layer **12** along its side edges **14**, **16**, **18**, **20**. The upper layer **28** may have a size, shape or configuration that generally corresponds to that of the lower layer **12**. This includes a base portion **38** that generally overlays the base portion **22** of the lower layer **12** and a backrest portion **40** that generally overlays the backrest portion **24** of the lower layer **12**. The upper layer **28** also tapers in width from the base portion **38** to a narrower end **42** that terminates at the top side edge **34** at the top of the backrest portion **40** to correspond to the taper of the lower layer **12**.

[0030] The materials of the upper layer **28** may be the same or different than those previously described for the lower layer **12**. In particular embodiments, however, the upper layer **28** may be formed from a different material. The upper layer **28** may be of an elastic material to allow the upper layer to stretch or expand, and contract. This may include a spandex material or spandex-containing materials. The spandex-containing materials may include fabric formed from blends of spandex with other materials, such as polyester, nylon, cotton, polypropylene, polyethylene, polyurethane, acrylic, vinyl (PVC), rayon, etc. In certain embodiments, the lower layer **12** may be formed from these elastic materials, as well. The upper layer **28** should have enough elasticity to allow the layer **28** to expand and contract for the purposes and uses described herein, such as conforming to seats of different sizes and shapes and to items that may be stored in the device **10**.

[0031] The upper layer may be liquid impermeable or resistant or may be liquid permeable. Because the lower layer **12** may have liquid impermeable or resistant characteristics, however, it may not be necessary for the upper layer **28** to have such characteristics to provide a liquid barrier to protect the seat top.

[0032] Furthermore, the upper layer or portions thereof may be opaque so that one cannot see through the material of the upper layer. This may obscure items that may be stored within the device **10** so that they cannot be seen from the exterior of the device **10**. In other instances, all or

portions of the upper layer **28** may be transparent or translucent.

[0033] In some embodiments, the lower layer **12** and/or upper layer **18**, or portions thereof, may be formed from one or more layers of material. In certain applications, an insulating material may be used for one or more layers forming the lower layer **12** and/or upper layer **18**, or portions thereof. The insulated layers **12** and/or **18** may allow the device **10** to be used for keeping items, such as food items, warm or cold while they are stored in the device **10**.

[0034] In the device **10**, the upper layer **28** has an opening **44**. The opening **44** is shown as being a transverse slit or slot formed in the upper layer **28** that may extend across all or a portion of the width of the upper layer **28**. The opening **44** may be located generally around or near the center of the upper layer **28** or formed in the base portion **38** or backrest portion **40** or near or at the junction of the base portion **38** and backrest portion **40**. The opening **44** is shown extending all the way to the right-side edge **32**. A closure device **46** is provided with the upper opening **44** for selectively opening and closing the upper opening **44**. The closure device **46** may include, but is not limited to, a zipper, a hook and loop fastener (e.g., Velcro®), snaps, buttons, buttons and loops, magnets, ties, toggles and cords, etc. In the embodiment shown, the closure device **46** constitutes a zipper assembly that opens the opening **44** from the right-side edge **32** towards the opposite left-side edge **30**.

[0035] Referring to FIG. 2, the upper layer **28** is joined around its perimeter along its side edges **30**, **32**, **34**, **36** to the perimeter of the lower layer **12** along side edges **14**, **16**, **18**, **20**, respectively, by seams **48**, **50**, **52**, **54**. The seams **48**, **50**, **52**, **54** may include permanent or non-permanent fastening methods. Examples of fastening methods for joining the layers together to form the seams include, but are not limited to, stitching, adhesive, glue, welding (e.g., ultrasonic, heat welding, etc.), double-sided tape, stapling, rivets, snaps, hook and loop fasteners, etc. Any or all of the seams **48**, **50**, **52**, **54** may be liquid tight so that liquids cannot seep through the seams. In other instances, any or all of the seams **48**, **50**, **52**, **54** may not be liquid tight.

[0036] In the embodiment shown, an extended bottom portion **56** that extends along the length of bottom side edge **20** of the base **22** of the lower layer **12** is folded upward to join the bottom side edge **36** of the upper layer **28**, where it is joined at line or seam **54**. Where the lower layer **12** is formed from a liquid impermeable or liquid resistant material, this facilitates catching and holding any liquid run off that would otherwise pass through the upper layer **28**, which may be formed from liquid permeable or resistant materials. In addition to the extended and folded bottom portion **56** of lower layer **12**, in certain embodiments, the lower layer **12** may be slightly larger or have a larger footprint than the upper layer **28** so that extended portions of the lower layer **12** are provided around all or a portion of its perimeter along the side edges **14**, **16** and top edge **18**. These extended areas may also be upturned and joined along the perimeter of the upper layer **28** so that the lower layer **12** has a bowl-like configuration. This also facilitates catching and holding any liquid run off that would otherwise pass through the upper layer **28**, which may be formed from liquid permeable materials.

[0037] The area between the lower and upper layers **12**, **28** form an interior space **58** (FIG. 3) of the storage device **10**. In some instances, the interior space **58** is a continuous interior space that extends the full or a major portion of the length and width of the device **10** between the lower and upper layers **12**, **28**. As used herein, “major portion” should be construed as being from 50% or greater. In some embodiments, the continuous interior space **58** may extend from at least, equal to, and/or between any two of 50 wt %, 51 wt %, 52 wt %, 53 wt %, 54 wt %, 55 wt %, 56 wt %, 57 wt %, 58 wt %, 59 wt %, 60 wt %, 61 wt %, 62 wt %, 63 wt %, 64 wt %, 65 wt %, 66 wt %, 67 wt %, 68 wt %, 69 wt %, 70 wt %, 71 wt %, 72 wt %, 73 wt %, 74 wt %, 75 wt %, 76 wt %, 77 wt %, 78 wt %, 79 wt %, 80 wt %, 81 wt %, 82 wt %, 83 wt %, 84 wt %, 85 wt %, 86 wt %, 87 wt %, 88 wt %, 89 wt %, 90 wt %, 92 wt %, 93 wt %, 94 wt %, 95 wt %, 96 wt %, 97 wt %, 98 wt %, 99 wt % and 100% of the length and width of the device **10**.

[0038] In certain embodiments, the seams **48**, **50**, **52**, **54** may close off the entire perimeter of the

layers **12**, **28**. In the embodiment shown, however, a portion of the right-side edge **32** of the upper layer **28** and a corresponding portion of the right-side edge **16** of the lower layer are not joined or coupled together along the full length of the side edges **14**, **30**. This area may be located at or around the middle of the device **10**. This facilitates forming a side opening **60** along a right-side edge of the device **10**. In many applications, the storage device **10** will be used on the passenger seat of vehicle. With a driver seated on the left side of the vehicle adjacent to the passenger seat, the driver can readily access the interior space **58** of the storage device **10** through this right-side opening **60**. In other embodiments, the opening **60** may be located on the left side of the device **10**, in instances where the driver seat may be located on the right side of the vehicle. The device **10** may also be used on back seats of the vehicle.

[0039] The upper opening **44** formed in the upper layer **28** may extend all the way to the right-side edge **32** so that it meets the side opening **60** in certain embodiments. This may allow an even larger opening to be formed in combination with the side opening **60** by opening the upper opening **44** to facilitate positioning of larger items or articles within the interior space **58**. In other embodiments, the upper opening **44** may not extend all the way to the side opening **60**.

[0040] Referring to FIGS. **1** and **2**, various securing devices are provided for securing the storage device **10** to the vehicle seat. Although the securing devices may be shown coupled to the upper layer **28** in FIG. **1**, they may also be coupled to the lower layer **12** or to both the lower and upper layers **12**, **28**. The securing devices may include side-securing straps **62**, **64** located on one or both sides of the device **10**, such as along the opposite side edges **14**, **16** of the upper layer **28**. The straps **62**, **64** are configured as flexible loops and may be of elastic or non-elastic material.

[0041] A headrest securing strap **66** is provided at the top end **68** of the of the device **10** for securing to a headrest of a vehicle seat. The securing strap **66** is sized and configured as a loop to loop over or pass around the seat headrest. The strap **66** may be formed from a single strap that loops over the headrest. Alternatively, the securing strap **66** may be formed from a pair of straps or strap segments that releasably couple together around the headrest, such as by buckles, quick-release buckles, D-rings, ties, hook and loop fasteners, ties, snaps, magnets, etc. The strap **66** may be formed from an elastic or non-elastic material.

[0042] A seat-base strap **70** is provided at the bottom **72** of the device **10**, as shown in FIG. **2**. The seat-base strap **70** is sized and configured to pass around the seat base or seat-base frame of the vehicle seat. In the embodiment shown, the strap **70** is coupled to opposite sides of the lower layer **12** at or near to where the extended portion **56** joins the rest of the lower layer **12**. The seat-base strap **70** is formed from a pair of strap segments **74**, **76** that releasably couple together by a buckle **78**, such as a quick-release buckle, or other fastening devices, such as those previously described for securing strap **66**. The strap segments **74**, **76** may be elastic or non-elastic. One or both of the strap segments **74**, **76** may be adjustable in length, such as with an adjustable buckle, to accommodate different size seat bases. Alternatively, or in addition to the adjustability, if the strap segments **74**, **76** are elastic, the elasticity of the segments **74**, **76** may allow sufficient expansion or contraction to accommodate different size seat bases.

[0043] The seat strap segments **74**, **76** are passed through the side straps **62**, **64**, respectively. The side straps **62**, **64** are not stationarily secured or coupled to the strap segments **74**, **76**, but are movable along the length of each segment **74**, **76**. This facilitates collapsing and deployment of the seat storage device **10**.

[0044] One or more additional storage compartments may also be provided on the device **10**. This may include one or more compartments **80**, such as pockets, coupled to the upper layer **28** in an area that is accessible during use. The compartment **80** may be formed from an additional layer of flexible material, such as fabric, that is coupled along its side edges to the upper surface of the upper layer **28** by various methods. This may include, but is not limited to, stitching, adhesive, glue, welding (e.g., ultrasonic, heat welding, etc.), double-sided tape, stapling, rivets, snaps, hook and loop fasteners, etc. Alternatively, an opening or slot may be formed in the upper layer **28** and a

container or pocket may be secured around the perimeter of such opening or slot to form the compartment **80**. In this configuration, items are passed through the opening or slot formed in the upper layer **28** and received in the secured compartment or pocket. The compartment **80** may be used for holding smaller items, such as mobile phones, pens or pencils, note pads, documents, etc. [0045] Referring to FIG. **4**, a vehicle seat **82** is shown with which the seat-top storage device **10** may be used. The vehicle seat **82** may be conventional in design and typically includes a seat base **84**, a backrest **86** that extends upward from the rear of the seat base **84**, and a headrest **88** that extends from the top of the backrest **86**. The headrest **88** may be adjustably mounted to the backrest **86** through one or more support members **90**, such as metal rods or bars, that allow the headrest **90** to raised, lowered or removed.

[0046] In use, the seat-top storage device **10** is secured to the seat **82** by passing or securing the headrest strap **66** over or around the headrest **88**. In some instances, it may be desirable to position the headrest strap **66** around the bottom of the headrest **88** or around the support members **90** where the headrest **88** is coupled to the backrest **86**.

[0047] The bottom portion **72** of the device **10** is positioned so that it overlays the front of the seat base **84** and extends across the width of the seat base **84**, as shown in FIG. **4**.

[0048] The two strap segments **74**, **76** of the seat-base strap **70** are passed through the side-securing straps **62**, **64**, respectively, along either side of the seat base **84**, or around the supporting structure or frame supporting the seat **82**, and secured together by buckle or other coupling device **66** to the rear of the seat **82**, as shown in FIG. **5**. The device **10** does not interfere with the seat controls or adjustment of the seat **82** and the seat can be used in a normal fashion. In other embodiments, the side-securing straps **62**, **64** may be secured to structures along the sides of the seat base **84**.

[0049] With the seat-top storage device **10** secured to the vehicle seat **82**, the seat-top storage device **10** may be in an initial closed and empty condition. In the closed condition, the opening **44** of the upper layer **28** is closed off by the closure device or zipper **46**, as shown in FIG. **6**.

[0050] To store items or articles within the interior **58** of the device **10** between the lower and upper layers **12**, **28**, in some instances, an item or article may be merely passed through the side opening **60** without having to open the upper opening **44**. This may be easily done by the driver with the use of one hand. If there is no side opening **60** or one wants to utilize the upper opening **44**, the upper opening **44** may be opened, as shown in FIG. **8**. Again, opening and closing of the zipper or closure device **46** may easily be accomplished by the driver with only one hand. The item or article **92** may then be positioned on upper surface of the lower layer **12** on the top of the seat base **84**. The upper opening **44** may then be closed by closing the closure device **46**.

[0051] Because the upper layer **28** may be formed from an elastic material, it can stretch to accommodate larger items or articles, as shown in FIG. **9**. Additional items or articles, such as mobile phone **96** may also be stored in the storage compartment **80**.

[0052] The seat-top storage device **10** also serves in protecting the vehicle seat **82** surface from dirt and spills. Because the lower layer **12** and/or the upper layer **28** may be formed from liquid impermeable or liquid resistant materials, spilled liquids are prevented from passing through the materials and onto the seat **82**. As shown in FIGS. **3** and **4**, where the upper layer **28** is liquid permeable and the lower layer **12** is liquid impermeable or resistant, if spilled liquids are spilled on the upper layer **20**, the liquids may pass through the upper layer **28** where they collect on the lower layer without passing to and damaging the surfaces of the vehicle seat **82**. The spilled liquids may then reside on the lower layer **12** until they can be removed. The extended portion **56** of the lower layer **12** that is folded upward to join the lower layer **28** forms a pocket that helps collect any spilled liquids flowing over the lower layer **12** that may flow to the front of the seat **82**. In embodiments where the lower layer **12** has a larger footprint than the upper layer **28** so that the side edges **14**, **16**, and optionally top edge **12**, are folded upward to form a bowl-like configuration, this also facilitates catching and holding liquids on the surface of the lower layer **12** so that they do not contact surfaces of the vehicle seat **82**.

[0053] When the device **10** no longer needs to be used on the vehicle seat **82**, a user can uncouple the headrest strap **66** from the headrest **88** and pull the top portion of the device **10** down while the seat-base strap **70** remains secured to the seat base **84**. The top unsecured portion of the device **10**, which may constitute a majority of the device **10**, can then be stored underneath the front of the vehicle seat **82**. The seat-base strap **70** does not need to be extremely tight and may have some degree of slack or elasticity around the seat base **84** as long as the strap **70** does not become detached or easily dislodged during use. Such slack or elasticity may facilitate the device **10** being stored under the seat **82** while still being secured to the seat base **84**. Because the side straps **62**, **64** are not stationarily secured or coupled to the strap segments **74**, **76**, they can move along the length of each seat-base strap segment **74**, **76** as it is being collapsed.

[0054] When it is desired to use the device **10** again, the user can merely pull the device **10** up from under the vehicle seat **82** so that it overlays the seat **82** and couple the headrest strap **66** to the headrest **88**, as previously described.

[0055] In other instances, the device **10** may be completely removed from the vehicle seat **82** by uncoupling the straps **66** and **70** so that the device **10** is completely uncoupled from the seat **82**. The device **10** may then be stored under vehicle seat **82**, the glove compartment, center console or other area of the vehicle. In certain embodiments, a storage container or case (not shown) may be provided with the device **10** for storage of the device. In some applications, a vehicle may be manufactured with seats configured to accommodate the storage device. In such instances, a storage area or compartment for storing the device **10** may be provided in the vehicle or vehicle seat itself for storing the device **10** when it is not in use.

[0056] Referring to FIG. **10**, another embodiment of a vehicle seat-top storage device **100** is shown. The device **100** is configured similarly to the device **10** and comprises a lower layer **102** of a flexible sheet material that is configured to overlay and conform to portions of the vehicle seat. As shown, the lower layer **102** is defined by left and right side edges **104**, **106** that are joined by top and bottom edges, **108**, **110**. The lower layer **102** includes a base portion **112**, configured to overlay a seat base of a vehicle seat, and a backrest portion **114**, configured to overlay a backrest of a vehicle seat. In certain embodiments, the lower layer **102** may taper in width along side edges **104**, **106** from an area of the base portion **112** to a narrower top end **118** of the backrest portion **114** that terminates at top edge **108**.

[0057] The materials of the lower layer **102** may be the same or similar to those of the lower layer **12** of the device **10**, previously described, and may have liquid impermeable or resistant characteristics, as well as the texturized, coated or be otherwise provided with a non-slip surface, such as those of the lower layer **12**.

[0058] The device **100** further comprises an upper layer **120** of a flexible sheet material that is defined by left and right side edges **122**, **124** that are joined by top and bottom edges **126**, **128**. The upper layer **120** may have a size, shape or configuration that generally corresponds to that of the lower layer **102**. This includes a base portion **130** that generally overlays the base portion **112** of the lower layer **102** and a backrest portion **132** that generally overlays the backrest portion **114** of the lower layer **102**. The upper layer **120** also tapers in width from the base portion **130** to a narrower end **134** that terminates at the top side edge **126** at the top of the backrest portion **132**.

[0059] The materials of the upper layer **120** may be the same or similar to the materials of the upper layer **28** of device **10**, previously described. This may include the elastic materials, liquid impermeable or liquid resistant materials, liquid permeable materials, opaque or non-see-through materials, transparent or translucent materials, as was described for the materials of the upper layer **28**.

[0060] The device **100** of FIGS. **11-17** differs from the device **10** in that includes a pair of left and right side gusset panels **136A**, **136B**, respectively, through which the lower and upper layers **102**, **120** are coupled together along their left and right side edges **104**, **122**, **106**, **124**. As shown in FIG. **10**, each of the gusset panels **136A**, **136B** has the same or a similar triangular shape or

configuration, although the gusset panels **136A**, **136B** can have other shapes, such as rectangular, quadrangular, polygonal, etc., as well. In the embodiment shown, the triangular gusset panels **136A**, **136B**, have an upper edge **138A**, **138B**, a back or rearward edge **140A**, **140B**, and a lower or bottom edge **142A**, **142B**. The upper edges **138A**, **138B** of the panels **136A**, **136B** form the longer or hypotenuse edge of the triangle and are joined to the left and right side edges **122**, **124**, respectively, of the upper panel **120**. This is generally along all or a portion of the backrest portion **132** of the upper layer **120**. In some embodiments, the upper edges **138A**, **138B** may extend further along all or a portion of the side edges of the base portion **130**, as well.

[0061] The rearward edges **140A**, **140B** of the panels **136A**, **136B** are joined along the side edges **104**, **106**, respectively, of the lower layer **102**. This is generally along all or a portion of the backrest portion **114** of the lower layer **102**. The bottom edges **142A**, **142B** are joined to the side edges **104**, **106** of the rear of the base portion **112** of the lower layer. The bottom edges **142A**, **142B** extend along only a portion of the length of the base portion **112** of the lower layer so that they terminate rearward of the forward edge **110** of the lower layer **102**. In such instances, the forward end of the base portion **112** of the lower layer **102** may be joined directly along the bottom edge **110** and side edges **104**, **106** to the bottom edge **128** and side edges **122**, **124** of the forward end of the base portion **130** of the upper layer **120**. In other embodiments, the bottom edges **142A**, **142B** of the gusset panels **136A**, **136B** are joined to the side edges **104**, **106** of the base portion **112** and extend the full length of the base portion **112**, terminating at the forward edge **110**. The edges of the various layers and/or panels may be joined by seams, such as those various seams **48**, **50**, **52**, **54**, as previously described for the device **100**.

[0062] In the embodiment shown, the upper edge **138B** of right gusset panel **136B** is joined to the side edge **124** of the upper layer **120** through a zipper assembly **144**. The zipper assembly **144** may extend along all or a portion of the length of the upper side edge **138B** of the gusset panel **136B** and be joined to the corresponding right side edge **124** of the upper layer **120**. In some embodiments, the zipper assembly **144** may further extend along the right side **124** of the upper layer where it is joined to the lower layer **102** so that the zipper assembly **144** extends along the entire length of the side edge **124** of the upper layer **120**.

[0063] The zipper assembly **144** allows the side edge **124** of the upper layer **120** to be selectively separated from the gusset panel **138B** and/or side edge **106** of lower layer **102** to form an opening **146**, as shown in FIG. **11**, to access an interior space **148** of the device **100** between the lower and upper layers **102**, **120**. In some instances, the interior space **148** is a continuous interior space that extends the full or a major portion of the length and width of the device **100** between the lower and upper layers **102**, **120**. In some embodiments, the continuous interior space **148** may extend from at least, equal to, and/or between any two of 50 wt %, 51 wt %, 52 wt %, 53 wt %, 54 wt %, 55 wt %, 56 wt %, 57 wt %, 58 wt %, 59 wt %, 60 wt %, 61 wt %, 62 wt %, 63 wt %, 64 wt %, 65 wt %, 66 wt %, 67 wt %, 68 wt %, 69 wt %, 70 wt %, 71 wt %, 72 wt %, 73 wt %, 74 wt %, 75 wt %, 76 wt %, 77 wt %, 78 wt %, 79 wt %, 80 wt %, 81 wt %, 82 wt %, 83 wt %, 84 wt %, 85 wt %, 86 wt %, 87 wt %, 88 wt %, 89 wt %, 90 wt %, 92 wt %, 93 wt %, 94 wt %, 95 wt %, 96 wt %, 97 wt %, 98 wt %, 99 wt % and 100% of the length and width of the device **100**.

[0064] In some embodiments, the lower layer **102**, upper layer **120**, and/or gusset panels **136A**, **136B**, or portions thereof, may be formed from one or more layers of material. In certain applications, an insulating material may be used for one or more layers forming the lower layer **102**, upper layer **120**, and/or gusset panels **136A**, **136B**, or portions thereof. The insulated layers may allow the device **100** to be used for keeping items, such as food items, warm or cold while they are stored in the device **100**.

[0065] With the device **100** employed with a vehicle passenger seat, the opening **146** is located along a right side edge of the device **100** that will typically be located on the driver's side so that the driver can access the opening **146**. In other embodiments, such as where the driver's seat is located on the right side of the vehicle, the opening **146** may be located on the left side of the

device **100**.

[0066] An additional upper opening **150** (FIG. **12**) is formed in the upper layer **120** to access the interior space **148**. The opening **150** may be the same or similar to the opening **44** of device **10**, extending transversely across all or a portion of the width of the upper layer **120**. In the embodiment shown, however, the opening **150** is formed as a longitudinal slit or slot that extends vertically or lengthwise along all or a portion of the length of the upper layer **120**. The opening **150** may be located generally around or near the center of the upper layer **120** or be formed in the base portion **130** or backrest portion **132** or near or at the junction of the base portion **130** and backrest portion **132**. The opening **44** may constitute a slit or slot formed in the material of the upper layer **120**. A closure device is provided with the upper opening **150** for selectively opening and closing the upper opening **150**. The closure device may include, but is not limited to, a zipper, a hook and loop fastener (e.g., Velcro®), snaps, buttons, buttons and loops, magnets, a tie, a toggle and cord, etc. In the embodiment shown, the closure device constitutes a series of longitudinally spaced apart buttons **152** located along one side of the opening **150** with corresponding spaced apart loops **154** located on the opposite side of the opening **150** that loop over the buttons **152** to secure and close the opening **150**. The loops **154** may be formed from an elastic material so they provide a degree of flexibility and can expand and contract.

[0067] Referring to FIG. **10**, various securing devices are provided for securing the storage device **100** to the vehicle seat. The securing devices may include side-securing straps **156**, **158** located on one or both sides of the device **100**. In the embodiment shown, these are coupled at one end to the bottom edges **142A**, **142B** of the gusset panels **136A**, **136B**, respectively, at or near the corner where the rearward edges **140A**, **140B** join or intersect the bottom edges **142A**, **142B**, respectively. The straps **156**, **158** may be coupled to other areas or components of the device **100**, however. The straps **156**, **158** may be of elastic or non-elastic material.

[0068] A headrest securing strap **160** is provided at the top end **162** of the of the device **100** for securing to a headrest of a vehicle seat. The securing strap **160** may be the same or similar to the securing strap **66** of the device **10**. The securing strap **160** is sized and configured to loop over or pass around the seat headrest and may be configured the same or similarly to the strap **60** of the device **10**, previously described. In the embodiment shown, the strap **160** is shown coupled to the top end **118** of the lower layer **102**, but may be coupled to the top end **134** of the upper layer **120** or both layers **102**, **120**. The securing strap **160** is sized and configured as a loop to loop over or pass around the seat headrest. The strap **160** may be formed from a single strap that loops over the headrest. Alternatively, the securing strap **160** may be formed from a pair of straps or strap segments that releasably couple together around the headrest, such as by buckles, quick-release buckles, D-rings, ties, hook and loop fasteners, ties, snaps, magnets, etc. The strap **160** may be formed from an elastic or non-elastic material.

[0069] The device **100** further comprises a seat-base strap assembly **164**. The seat-base strap assembly **164** includes a seat-base apron **166** that is formed from a flexible layer of material configured to extend across the front of the vehicle seat base. The material of the apron **166** may be the same or similar to the materials forming the lower and upper layers **102**, **120**, and may be elastic or non-elastic. The apron **166** may be a separate piece of material or may be formed from an extended portion of either of the layers **102**, **120**. In the embodiment shown, the apron **166** is a separate piece of material that is joined at its upper edge **168** along the forward edge of the device **100** where the forward edges **110**, **128** of the lower and upper layers **102**, **120** are both joined together. Left and right side portions **170**, **172** of the apron **166** are configured to extend around either side of the front of the seat base a distance and are joined along the sides of the device **100** where the lower and upper layers **102**, **120** are joined along their side edges near the forward end **174** of the device **100**.

[0070] The seat-base strap assembly **164** further comprises a seat-base strap **176**. The seat-base strap **176** may be configured the same or similar to the seat-base strap **70** of the device **10** and may

be formed by a pair of strap segments **178, 180**. The strap segments **178, 180** are coupled at one end to the left and right side portions **170, 172**, respectively, of the bottom of the apron **166**. The seat-base strap **176** is sized and configured to pass around the seat base or seat-base frame of the vehicle seat. The strap segments **178, 180** releasably couple together by a buckle assembly **182**, such as a quick-release buckle, or other fastening devices, such as those previously described. The strap segments **178, 180** may be elastic or non-elastic. One or both of the strap segments **178, 180** may be adjustable in length, such as with an adjustable buckle, to accommodate different size seat bases. Alternatively, or in addition to the adjustability, if the strap segments **178, 180** are elastic, the elasticity of the segments **178, 180** may allow sufficient expansion or contraction to accommodate different size seat bases.

[0071] The seat strap segments **178, 180** are passed through the side straps **156, 158**, respectively. In some embodiments, the side straps **156, 158** are not stationarily secured or coupled to the strap segments **178, 180**, but are movable along the length of each segment **178, 180** to different positions. This facilitates collapsing and deployment of the seat storage device **100**. In other embodiments, the side straps **156, 158** may be stationarily coupled to the seat strap segments **178, 180**.

[0072] The device **100** may further comprise a pocket or compartment **184**, which may be configured the same or similarly to the compartment **80** of device **10**. This may be used for storing various items, such as the mobile phone **96** (FIG. **13**).

[0073] Referring to FIG. **11**, the device **100** is shown employed with the vehicle seat **82**, which was described previously with respect to the device **10**. In use, the seat-top storage device **100** is secured to the seat **82** by passing or securing the headrest strap **160** over or around the headrest **88**. In some instances, it may be desirable to position the headrest strap **160** around the bottom of the headrest **88** or around the support members **90** (FIG. **13**) where the headrest **88** is coupled to the backrest **86**.

[0074] The seat-base apron **166** of the device **10** is positioned so that it overlays the front of the seat base **84** and extends across the width of the seat base **84**, as shown in FIG. **11**. The two strap segments **178, 180** pass through the left and right side-securing straps **156, 158**, respectively, along either side of the seat base **84**, or around the supporting structure or frame supporting the seat **82**, and are secured together by the buckle assembly **182** or other coupling device to the rear of the seat **82**. As shown, the seat-strap segments **178, 180** are spaced apart from the upper surface of the seat base **84** of the vehicle seat **82** when secured around the seat base **84** so that the side straps **156, 158** are pulled downward and may be in tension to facilitate holding the device **100** securely on the seat base **84**. The side straps **156, 158** may be positioned on the device **100** so that they are located along the seat base **84** and may be located at or near the junction where the seat base **84** and the backrest **86** of the vehicle seat **82** meet, in certain instances.

[0075] With the seat-top storage device **100** secured to the vehicle seat **82**, the seat-top storage device **100** may be in an initial closed and empty condition. In the closed condition, the side opening **146** is closed off by the closure device or zipper **144**, as shown in FIG. **12**.

[0076] To store items or articles within the interior **148** of the device **100** between the lower and upper layers **102, 120**, the zipper or closure **144** is opened and items or articles, such as the camera **186**, handbag or purse **188**, and tablet **190** (FIG. **13**), are passed through the side opening **146**. The items or articles **186, 188, 190** may then be positioned on the upper surface of the lower layer **102** on the top of the seat base **84**. The side opening **146** may then be closed by closing the zipper or closure device **144**. Opening and closing of the side opening **146** can easily be done by the driver with one hand.

[0077] The use of the gusset panels **136A, 136B** provides a greater volume to the interior **148** and of the device **100** and facilitates the base portion **112** of the lower layer **102** reaching to the back of the seat base **84** to form a deep pocket. Additionally, because the upper layer **120** may be formed from an elastic material, it may stretch to accommodate larger items or articles positioned within

the interior of the device **100**.

[0078] In addition to the side opening **146**, items or articles may also be placed through the upper opening **150** formed in the upper layer **120**, which may be closed and secured with the closures **152**, **154**. Again, the closures used for the upper opening **150** may be easily manipulated by a driver with the use of one hand.

[0079] In addition to storing items and articles, the device **100** may be used for securing a pet or animal (e.g., dog, cat, etc.) on the vehicle seat **82**. Referring to FIG. **14**, a pet **192** is shown positioned with its body and legs positioned within the interior **148** of the device **100**, while its head and neck project out of the opening **150**. The buttons **152** and elastic loops **154** facilitating selective partial closing of the opening **150** to a degree sufficient to retain the animal within the device **100**, while allowing it to comfortably extend its head and neck through the opening **150**. The use of the button and elastic loop closure allows some degree of expansion to give the animal sufficient room to move and also prevents the animal's hair from catching, which may occur with other types of closure devices. The non-slip upper surface of the lower layer **102** also helps the pet, as well as articles, from sliding on the upper surface of the lower layer **102**.

[0080] As with the device **10**, the seat-top storage device **100** also serves in protecting the vehicle seat **82** surface from dirt and spills.

[0081] Referring to FIGS. **15** and **16**, when the device **100** no longer needs to be used on the vehicle seat **82** or needs to be removed to accommodate a passenger, a user can merely uncouple the headrest strap **160** from the headrest **88**. This can be readily done with one hand by the driver. With the seat-base strap **176** still secured around the seat base **84**, the device **100** can be folded, gathered or rolled downward in a collapsed configuration, as shown in FIG. **15**, and conveniently stored under the seat **82** in lower space **194** between the seat supporting structures **196**, **198** (FIG. **11**) so that it is out of the way and out of sight. If the collapsed device **100** has a large width, the ends of the collapsed device **100** can be folded inward for storage, as shown in FIG. **16**. The seat-base strap **176** does not need to be extremely tight and may have some degree of slack around the seat base **84** so long as the strap **176** does not become detached or easily dislodged during use. Such slack may facilitate the device **100** being stored under the seat **82** while still remaining secured to the seat base **84**.

[0082] When the device **100** is stored, it is out of the way so that it does not touch or interfere with a passenger sitting on the vehicle seat **82**. The device remains out of sight and is not visible when stored under the seat or other storage area.

[0083] Referring to FIG. **17**, in some embodiments, a storage area or compartment or drawer **200** for storing the device **100** may be provided with the device **100** or with the vehicle or vehicle seat itself for storing the device **10** when it is not in use. In some applications, vehicles may be manufactured and provided with the storage device seat-top storage devices described herein already installed. In such cases, the vehicle may be provided with a storage compartment or drawer, such as the compartment or drawer **200**. The compartment or drawer may be movable and may be mounted on tracks, rails or other devices (not shown), which allows it to be readily pulled out or extended from under the seat **82** to store the device **100** and then be pushed back or retracted under the vehicle seat **82** so that it is out of the way and out of sight. The compartment or drawer **200** may be located forward of the seat base **84** of the vehicle seat **82** when extended out. In some embodiments, a movable cover or lid **202** may be used to selectively open or close the compartment or drawer **200** so that the device **100** is covered. In other embodiments, no lid is provided. The compartment or drawer **200** may be provided with the device **100** and/or provided as part of the vehicle seat **82** itself and be configured to hold and store the collapsed device **100**.

[0084] FIG. **18** shows still another embodiment of a seat-top storage device **204**. The device **204** is similarly constructed to the device **100**, previously described, with similar components labeled with the same reference numerals. The device **204** of FIG. **18** differs from device **100** in that it is shown with a zipper assembly **206** as the closure device for opening **150**.

[0085] The zipper assembly **206** is shown in more detail in FIG. **19**. The zipper assembly **206** may include strips of zipper tape **208** coupled along either side of material of the upper layer **120** surrounding the opening **150**. The strips of zipper tape **208** extend along the length of the opening **150**, with the ends of the zipper tape **208** being joined together at or near the ends of the opening **150**. Each strip of zipper tape **208** has a series or chain of zipper teeth **210** that extends along the length of the tape **208** that are configured to interlock with corresponding chain of zipper teeth **210** of the other tape strip **208** located on the opposite side of the opening **150**. The teeth **208** may be of a metal (e.g., brass, aluminum, etc.) or a polymeric or plastic material, such as nylon.

[0086] The zipper assembly **206** includes one or more sliders **212** that are movable along the chain of zipper teeth **210** of each strip of zipper tape **208** to selectively close or interlock the teeth **210** to close the opening **150** and to open or unlock the teeth from one another to open the opening **150**. In the embodiment shown, there are two opposite sliders **212** that are movable along the length of the zipper assembly **206**. This allows the size of the opening formed by the zipper assembly **206** to be adjusted and also allows the position of the opening to be moved to selected positions between the sliders **212** along the length of the zipper assembly **206**. Stops (not shown) may be provided at each end of the zipper assembly **204** to prevent the sliders **212** from running off or detaching from the zipper assembly **204**.

[0087] Pull tabs **214** are provided with each of the sliders **212** that are configured to be manually grasped to allow a user to adjust the position of each sliders **212** to selectively open or close the opening formed by the zipper assembly **204**.

[0088] In some applications, the zipper assembly **206** is a non-slip zipper assembly that is configured to prevent the zipper assembly **206** from unintentionally opening or closing, such as by pulling apart the material surrounding the opening **150** on either side of the zipper assembly **206**. This may be provided by using a locking zipper slider for the sliders **212**. Such locking sliders are configured to engage the zipper teeth to prevent the slider **212** from moving in one or both directions along the length of the zipper assembly **206** unless force is applied to the slider itself so that it is moved, such as by pulling on the pull tabs **214**. This prevents the inadvertent opening or closing of the device opening **150**. This may be important in such instances where the device **204** is used to secure and retain a pet within the device **204**. The non-slip zipper assembly prevents the pet from forcing the zipper assembly open so that the pet is retained by the device **204** and it is unable to expand the opening **150**.

[0089] A zipper guard **216** is also provided with the device **204**. The zipper guard **216** is comprised from a flap or layer of material, which may be the same or different from the material of the upper layer **120**, that extends along and overlays the chain of zipper teeth **210** on each side of the zipper assembly **206**. The material of the zipper guard **216** may extend beyond the edges of the chain of zipper teeth **210** to provide protection from the edges of the zipper teeth **210** and to prevent items, such as hair, fur, clothing, etc., from getting caught in the zipper assembly **206**.

[0090] The seat-top storage devices as described herein provide a secure, convenient and easy-to-use means for storing various items or articles, such as electronic devices, mobile phones, tablets, laptop computers, bags or purses, shopping bags, clothing items, documents, food items and snacks, or other personal items. The devices may also be used to comfortably secure pets or animals on a vehicle seat, while allowing them to see out and giving them a degree of freedom. The device is easy to install and use and is readily accessible from the driver's seat. A driver or user can readily access, erect, collapse and store the device with one hand and minimal effort. The device also helps protect the vehicle seat from dirt and spills. Furthermore, it may be formed from opaque materials to obscure or conceal articles stored within the device. This may facilitate protecting the articles from damaging sunlight as well as help deter theft that might otherwise occur should the stored items or articles be visible through the vehicle windows.

[0091] While the invention has been shown in some of its forms, it should be apparent to those skilled in the art that it is not so limited, but is susceptible to various changes and modifications

without departing from the scope of the invention. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the invention.

Claims

1. A vehicle seat-top storage device configured to secure to a vehicle seat top, the vehicle seat-top storage device comprising: a lower layer formed from a flexible material configured to overlay and conform to portions of a vehicle seat having a seat base and a backrest that extends upward from the seat base; an upper layer formed from a flexible sheet material coupled along side edges to the lower layer to define an interior space that extends across a major portion of the length and width of the device between the lower and upper layers; and an opening formed in the upper layer that allows access to the interior space through the opening, the opening being a slit or slot located at or near the center of the upper layer that extends longitudinally along all or a portion of the length of the upper layer, the opening being provided with a closure device for selectively opening and closing the opening.
2. The vehicle seat-top storage device of claim 1, wherein: all or portions of the lower layer are formed from a liquid impervious or liquid resistant material to prevent liquids from passing through the lower layer.
3. The vehicle seat-top storage device of claim 1, wherein: the lower layer has a backrest portion configured to overlay the backrest and a base portion configured to overlay the seat base of the vehicle seat, the lower layer extending from the backrest portion to the front of the seat base when the storage device is secured to and positioned over the vehicle seat top.
4. The vehicle seat-top storage device of claim 1, further comprising: at least one storage compartment provided on the upper layer for storing items on the upper layer.
5. The vehicle seat-top storage device of claim 1, further comprising: a second opening is provided in the device along a side edge of the device to allow access to the interior of the device through the side.
6. The vehicle seat-top storage device of claim 1, wherein: the storage device covers a majority of the surface of the upper surface of the seat base and backrest.
7. The vehicle seat-top storage device of claim 1, wherein: the closure device is at least one of a zipper assembly, a hook and loop fastener, snaps, buttons, buttons and loops, magnets, a tie, and a toggle and cord.
8. The vehicle seat-top storage device of claim 1, wherein: at least one of the upper and layers, or portions thereof, is formed from an insulating material.
9. The vehicle seat-top storage device of claim 1, wherein: the closure device is a zipper assembly having two opposite sliders mounted on a zipper chain of the zipper assembly.
10. The vehicle seat-top storage device of claim 1, wherein: the closure device is a non-slip zipper assembly that comprises a locking slider.
11. The vehicle seat-top storage device of claim 1, wherein: the closure device is a zipper assembly having zipper teeth provided on either side of the opening, and wherein a zipper guard extends over the zipper teeth to prevent items from being caught in the zipper teeth.
12. The vehicle seat-top storage device of claim 1, wherein: the upper and lower layers are coupled together along at least a portion of the side edges of the upper and lower layers by a gusset panel.
13. The vehicle seat-top storage device of claim 1, wherein: all or a portion the lower layer is provided or formed with a material with a non-slip surface.
14. The vehicle seat-top storage device of claim 1, further comprising: a seat-base apron that is formed from a flexible layer of material configured to extend across the front of the vehicle seat base when the storage device is secured to and positioned over the vehicle seat top.
15. A vehicle seat-top storage device configured to secure to a vehicle seat top, the vehicle seat-top storage device comprising: a lower layer formed from a flexible material configured to overlay and

conform to portions of a vehicle seat having a seat base and a backrest that extends upward from the seat base; an upper layer formed from a flexible sheet material coupled along side edges to the lower layer to define an interior space that extends across a major portion of the length and width of the device between the lower and upper layers; an upper opening formed in the upper layer that allows access to the interior space through the upper opening, the upper opening being a slit or slot located at or near the center of the upper layer that extends longitudinally along all or a portion of the length of the upper layer, the upper opening being provided with a closure device for selectively opening and closing the upper opening; a side opening located along a side of the device that allows access to the interior space through the side opening; and a securing device for securing the storage device to the seat base.

16. The vehicle seat-top storage device of claim 15, wherein: the closure device is a zipper assembly having two opposite sliders mounted on a zipper chain of the zipper assembly.

17. The vehicle seat-top storage device of claim 15, wherein: the closure device is a non-slip zipper assembly that comprises a locking slider.

18. The vehicle seat-top storage device of claim 15, wherein: the closure device is a zipper assembly having zipper teeth provided on either side of the opening, and wherein a zipper guard extends over the zipper teeth to prevent items from being caught in the zipper teeth.

19. The vehicle seat-top storage device of claim 15, wherein: the upper and lower layers are coupled together along at least a portion of the side edges of the upper and lower layers by a gusset panel.

20. A method of storing items in a vehicle seat-top storage device, the method comprising: providing a vehicle seat-top storage device comprising: a vehicle seat-top storage device configured to secure to a vehicle seat top, the vehicle seat-top storage device comprising: a lower layer formed from a flexible material configured to overlay and conform to portions of a vehicle seat having a seat base and a backrest that extends upward from the seat base; an upper layer formed from a flexible sheet material coupled along side edges to the lower layer to define an interior space that extends across a major portion of the length and width of the device between the lower and upper layers; an opening formed in the upper layer that allows access to the interior space through the opening, the opening being a slit or slot located at or near the center of the upper layer that extends longitudinally along all or a portion of the length of the upper layer, the opening being provided with a closure device for selectively opening and closing the opening; securing the vehicle seat-top storage device to a vehicle seat; and positioning an item to be stored within the interior space between the lower and upper layers while the vehicle seat-top storage device is secured to the vehicle seat.
